Vrishank Chandrasekhar

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Education

Massachusetts Institute of Technology, Biological Engineering, minor in Computer Science — Cambridge, MA, Aug 2025 – June 2029

- Activities and Societies: Y Combinator Startup School 2026, NEET Living Machines, MIT Associate Advisor, MIT BioMakers Associate, Genetics Tutor, MIT Undergraduate Research Technology Conference Executive & Webmaster, Co-President of MIT Ohms: South Asian Acappella

Lynbrook High School, San Jose, California — San Jose, CA, Aug 2021 – June 2025

- Activities: Machine Learning Club (President), Research Club (Vice President), Chamber Orchestra

West Valley College — Saratoga, CA, June 2024 – May 2025

Experience

Aghaeepour Lab | Stanford Biomedical Data Science, Undergraduate Research Intern — Stanford, CA, July 2026 – present, 1 month

Preparing multimodal data pipelines to support pre-training and fine-tuning of RL-based foundation models for clinical reasoning and neonatal patient trajectory prediction. Supervised by Dr. Weijieying Ren.

Vázquez-García Lab | Broad Institute of MIT and Harvard, Undergraduate Researcher — Cambridge, MA, Sept 2025 – Aug 2026, 1 year

Supervised by Prof. Ignacio Vázquez-García (Broad Institute, Massachusetts General Hospital, Harvard Medical School). Deploying single-cell genomics data analysis pipelines to study tumor clonal evolution.

Rabinovitch Lab | Stanford University School of Medicine, Paid Bioinformatics and Cardiovascular Research Intern — Stanford, CA, June 2023 – Aug 2025, 2 years 3 months

Applied bioinformatics, transcriptomics, imaging informatics, and functional assays to investigate how loss-of-function mutations can cause the pathogenesis of pulmonary arterial hypertension (PAH) and other rare cardiovascular diseases.

- Developed a computational framework to analyze RNA-seq data from co-cultures of wild-type and knockout stem cell-derived endothelial and smooth muscle cells; optimized co-cultures for experimental consistency.
- Investigated the role of TBX4 loss in driving endothelial-like gene expression in smooth muscle cells, with broader contributions to research on promoter optimization for haploinsufficiency research.
- Selected as the first high school student to participate in the Single-Cell Omics Analysis Jamboree hosted by the IGVF Consortium. Collaborated with other laboratories to test and analyze single-cell epigenetic data using a novel analysis pipeline developed by the Broad Institute.

Lynbrook High School, Independent Biomedical Informatics Researcher — San Jose, CA

Developing machine learning algorithms using electronic health records (EHRs) and medical imaging to predict adverse clinical events. — Mar 2022 – June 2025, 3 years 4 months

- Developing supervised, weakly supervised, self-supervised, and unsupervised machine learning and deep learning models for medical image analysis and clinical natural language processing.
- Grand Prize Winner of the Synopsys Silicon Valley Science and Technology Championship and 1st Grand Award and Top Award winner at the Regeneron International Science and Engineering Fair. Named a 2025 Regeneron Science Talent Search Scholar. Presented at SIYSS and attended Nobel Week festivities.

Skills

Dry Lab

Wet Lab

Projects

Discovery of a Novel Digital Biomarker for Early Pan-Cancer Survival and Recurrence Prediction via Free-Text Clinical Narratives

Used natural language processing to assess the use of free-text clinical narratives in fitting survival and recurrence outcomes for pan-cancer patients at initial diagnosis, enabling risk stratification across 32 cancer types. Earned top awards at 2025 International Science and Engineering Fair. Named a Top-300 Scholar in the 2025 Regeneron Science Talent Search (STS).

Investigating the Role of TBX4 Loss-of-Function in Pulmonary Arterial Hypertension (PAH)

Developed an automated RNA-sequencing differential expression and functional analysis application, using co-culture models of stem-cell differentiated wild-type and mutated endothelial and smooth muscle cells, to study how TBX4 loss-of-function mutations can lead to a PAH phenotype.

Awards

MIT Huang-Hobbs Innovation Award — 2025

Awarded Grand Prize for my research at the MIT BioMakers: From Bench to Body Symposium. Judged and recognized by MIT Biological and Chemical Engineering faculty and Broad Institute Core Faculty, this award honors an exceptional MIT student whose pioneering work demonstrates creativity, technical excellence, and real-world impact, and pushes boundaries in human health, medicine, and life sciences innovation to advance greater societal impact and accessibility.
MIT BioMakers: From Bench to Body Symposium

Dudley R. Herschbach SIYSS Top Award — 2025

Awarded 1 of 12 best-in-fair Top Awards out of 1,657 finalists and 10 million global entries at the 2025 Regeneron International Science and Engineering Fair (ISEF). Received a free trip to present research at the Stockholm International Youth Science Seminar; served as a panelist in "Minds of the Future" with Prof. Omar Yaghi (2025 Nobel Laureate in Chemistry). Attended the Nobel Prize Lectures & Ceremonies. Met the 2025 Laureates at the Nobel Reception & the Swedish U.S. Embassy.
Regeneron International Science and Engineering Fair (ISEF)

First Grand Award and Best of Category in Translational Medical Science at ISEF ‘25 2025

Selected as Best of Category in Translational Medical Science worldwide for my research discovering a low-cost, non-invasive, text-based digital biomarker for pan-cancer prognosis prediction via free-text clinical narratives. $6,000 prize.
Regeneron International Science and Engineering Fair (ISEF)

2025 Regeneron Science Talent Search (STS) Scholar 2025

Named a Top-300 Scholar in the nation’s oldest and most prestigious science and mathematics research competition for high school seniors. $2,000 prize.
Society for Science

2nd Place Prize by Silicon Valley Engineering Council

$3,000 Cash Prize issued by SVEC, dedicated to promoting STEM innovation by fostering curiosity and technical excellence in future engineers and scientists.
Silicon Valley Engineering Council (SVEC)

Grand Prize Winner at Synopsys Silicon Valley Science and Technology Championship

Selected to compete at the Regeneron International Science and Engineering Fair (ISEF), the world’s largest and most prestigious international pre-college STEM research competition, as an ISEF finalist from the Santa Clara Valley Science and Engineering Fair, the most competitive regional ISEF-affiliated fair in the nation.
Synopsys Silicon Valley Science and Technology Championship

Conference Presentations

Stockholm International Youth Science Seminar (SIYSS)

Invited Research Presenter and Panelist with Prof. Omar Yaghi (2025 Nobel Laureate in Chemistry)

Nobel Lectures Week, Nobel Prize Ceremony and Banquet in Stockholm, Sweden

Attended

American Thoracic Society

Linking Novel Extra-Thoracic Disease Mechanisms with PAH

IEEE MIT Undergraduate Research Technology Conference

Oral presentation

West Coast Biological Sciences Undergraduate Research Conference

Poster presentation

Stanford Cardiovascular Research Symposium

Poster presentation (4 times)

Maternal and Child Health Research Institute (MCHRI) Symposium

Poster presentation

2023 American Chemical Society National Meeting

Poster presentation